

DR. BABASAHEB AMBEDKAR TECHNOLOGICAL UNIVERSITY, LONERE					
Regular/Supplementary Winter Examination – 2025					
Course: Mechanical Engineering		Branch: Mechanical Engineering		Semester: VI	
Subject Code & Name: BTMEC605C_Y19 Renewable Energy Sources					
Max Marks: 60		Date:10/07/2025		Duration: 3 Hr.	
Instructions to the Students: 1. Each question carries 12 marks. 2. Question No. 1 will be compulsory and include objective-type questions. 3. Candidates are required to attempt any four questions from Question No. 2 to Question No. 6. 4. The level of question/expected answer as per OBE or the Course Outcome (CO) on which the question is based is mentioned in () in front of the question. 5. Use of non-programmable scientific calculators is allowed. 6. Assume suitable data wherever necessary and mention it clearly.					
				(Level/CO)	Marks
Q. 1	Objective type questions. (Compulsory Question)				12
1	Which of the following is considered a non-renewable energy resource?			Knowledge	1
	a. Solar energy	b. Wind energy	C.Coal d. Biomass		
2	Which energy resource has the largest proven reserves in India?			Knowledge	1
	a. Coal	b. Natural Gas	c. Uranium d. Crude Oil		
3	Nuclear fission involves:			Knowledge	1
	a.Splitting of a heavy nucleus into lighter nuclei	b. Combining of light nuclei into a heavier nucleus	c. Conversion of solar energy into electricity d. None of the above		
4	Which instrument is commonly used to measure solar radiation?			Knowledge	1
	a. Pyranometer	b. Anemometer	c. Hygrometer d. Thermometer		
5	Which material is commonly used for the absorber plate in flat plate solar collectors?			Understanding	1
	a. Aluminum	b. Copper	c. Plastic d. Stainless steel		
6	Which of the following is an advantage of solar energy?			Understanding	1
	a. Emits greenhouse gases	b. Limited availability during day	c. Renewable and non-polluting d. High operational cost		
7	What is the role of tracking systems in concentrating collectors?			Understanding	1
	a. To reduce heat loss during operation	b. To maintain alignment with the sun's position	c. To circulate the working fluid d. To measure solar radiation		
8	What is the most common application of solar water heating systems?			Application	1

	a. Industrial steam generation	b. Domestic hot water supply	c. Air conditioning	d. Electricity generation		
9	Flat plate collectors are typically tested for:				Application	1
	a. Weight and durability	b. Electrical conductivity	c. Thermal performance and efficiency	d. Noise level		
10	What is the primary source of wind energy?				Application	1
	Ocean currents	Rotation of the Earth	Uneven heating of the Earth's surface by the sun	Gravitational pull of the moon		
11	What is the primary purpose of a bio-conversion plant?				Application	1
	a. Generate solar power	b. Convert biomass into usable energy	c. Store wind energy	d. Extract fossil fuels		
12	What is the source of geothermal energy?				Analysis	1
	a. Heat from the sun	b. Heat generated by the Earth's core	c. Combustion of fossil fuels	d. Chemical reactions in the ocean		
Q. 2	Solve the following.					12
A)	Enlist various types of energy resources available and recommend future energy resource.				Application	6
B)	What are energy resources? Discuss the current status of energy conversion in India.				Understand	6
Q.3	Solve the following.					12
A)	What are the different types of concentrators used in concentrating solar collectors? Describe each briefly.				Remember	6
B)	What are the common methods used for measuring solar radiation?				Understand	6
Q. 4	Solve Any Two of the following.					12
A)	Explain flat plate solar collectors and describe the selection criteria for choosing suitable flat plate collectors.				Understand	6
B)	How are flat plate solar collectors tested to evaluate their performance and efficiency?				Understand	6
C)	What are the basic components of a solar water heating system, and what is the working principle behind it?				Apply	6
Q.5	Solve Any Two of the following.					12
A)	State the key advantages of wind energy as a renewable energy source.				Apply	6
B)	Define biomass and explain its various forms with suitable examples.				Understand	6

C)	List and explain the different types of windmills in detail.	Understand	6
Q. 6	Solve Any Two of the following.		12
A)	What is tidal energy? Explain the process of its generation.	Understand	6
B)	What is geothermal energy? Describe how it is harnessed for electricity production.	Apply	6
C)	What is hydroelectric energy? Explain the method of its generation.	Understand	6
	*** End ***		